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August 2, 2026

The Editors
ACS Omega

Dear Editors,

I am submitting the manuscript “**Fully Symmetry-Conditioned Rigid-Body Flow Matching for Molecular-Crystal Structure Prediction**” for consideration as a Research Article in *ACS Omega*. This manuscript is transferred from the *Journal of Chemical Information and Modeling* (manuscript ID [JCIM MS ID]) at the Editor’s invitation following the decision of [DECISION DATE].

Rigid-body flow matching has become an attractive route to molecular-crystal structure prediction: freezing each molecule’s conformer reduces the problem to a lattice, a set of fractional centroids, and a per-molecule orientation on $SO(3)$, a low-dimensional target that fast flow-matching samplers solve in tens of steps. Yet the current generation of rigid-body molecular-crystal generators leaves crystallographic symmetry unconditioned on the orientation degree of freedom, and symmetry- or Wyckoff-conditioned generation has to date been demonstrated only for inorganic atomic sites. This manuscript closes that gap and makes the conditioning deployable.

The contributions are:

- **A mechanism, then a method.** The per-molecule orientation target factorizes into a space-group-determined relative rotation, which a flow learns, and a gauge-free asymmetric-unit pose, which it cannot regress from packing. This explains why an unconditioned orientation flow stalls, and it identifies exactly what to condition on.
- **Deployable symmetry-coset conditioning.** A leak-free coset label — the generating space-group operation, recovered from centroids against a symmetry template at sampling time, never from the observed rotation — conditions the orientation flow. It recovers two-thirds of an oracle-codebook benefit (+41.1% held-out non-reference orientation loss vs. +27.5% for a paired control), and an $SO(3)$ -averaged objective closes the rest (+47.7%). Supplying the coset at generation collapses the end-to-end median orientation error sevenfold ($93.5^\circ \rightarrow 13.4^\circ$), and the advantage widens with data.
- **The first fully symmetry-conditioned molecular-crystal flow.** Extending the conditioning to the lattice — a crystal-family mask on an $O(3)$ -invariant log-metric cell parametrization — composes with the orientation coset to condition every symmetry-constrained degree of freedom. Paired with an unsupervised, symmetry-preserving packing finisher and a fully ablated stack of levers, it lifts held-out exact match from 0% to 6.9% (strict best-of-10, StructureMatcher stol 1.0) — statistical parity with the symmetry-free MolCrystalFlow ($\sim 8\%$), from a flow that matched nothing — and 10.7% with orientation test-time augmentation.

I believe the work fits *ACS Omega*’s interdisciplinary remit. It sits at the interface of crystallography, machine learning, and solid-state chemistry: the result is a design rule for chemists building molecular-crystal structure generators — condition orientation on the generating coset and the lattice on the crystal family, sample only the free pose, and finish with a symmetry-preserving physical relaxation — reported with each lever’s contribution quantified rather than as a single state-of-the-art number. The limitations are stated explicitly, including the comparable-task (not head-to-head) nature of the MolCrystalFlow comparison and the confidence interval on the parity claim.

For the transfer I have restructured the manuscript to *ACS Omega*’s section order, placing Methods after the Conclusions, and have made no changes to the data, analysis, or claims.

I confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. It has a single author with no competing financial interest to declare. The source code, the CSD refcode manifest, and the export/factorization pipeline are openly available (<https://github.com/fronkt/symmc-flow>; archived at Zenodo, <https://doi.org/10.5281/zenodo.21500734>), so every reported number regenerates from the deposited artifacts. The Cambridge Structural Database is licence-restricted; the underlying structures are not redistributed, but the deposited refcode manifest reproduces the corpus from a licensed CSD installation.

Thank you for considering this work.

Sincerely,

Frank Cai
Purdue University